

Chapter 6 Fluid resistance and energy loss

Section 1 Overview of flow resistance and energy loss

Section 2 Two regimes/states of fluid motion

Section 3 Laminar flow motion in circular tube

Section 4 Turbulent flow motion

Section 5 Frictional head loss of turbulent flow in
circular tubes

Section 6 Frictional head loss of turbulent flow in
noncircular tubes

Section 7 Minor head loss

Section 8 Concept of boundary layer

Section 9 Drag due to flow around a body

Chapter 6 Fluid resistance and energy loss

Key study points:

- Classifications of flow resistance and head loss
- Two status of viscous flow: laminar flow and turbulence flow
- Relationship between frictional head loss and shear stress; shear force distribution on the flow cross section of circular pipe
- Laminar flow in circular pipe: velocity distribution; relationship between head loss and mean velocity; relationship between frictional loss coefficient and Re
- Turbulent flow: turbulent fluctuation and its time average; Reynolds stress; mixing-length theory
- Frictional loss of turbulent flow: Nikuradse's Experiment; Moody's diagram; empirical formulations; Frictional loss of non-circular pipes
- Minor head losses: those for sudden enlargement/contraction sections; minor head loss factor
- Boundary layer: concepts; separation of boundary layer; resistance force of flow around a body

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-  1. Classifications of flow resistance and head loss
- 2. Basic formula for head loss calculation
- 3. Basic equation for uniform flow
- 4. Friction velocity; Shear stress distribution in circular tube

Section 1 Overview of flow resistance and head loss

Origin of flow resistance and energy loss: fluid viscosity and turbulence.

1. Classifications of flow resistance and head loss: Two forms

(1) Frictional resistance (沿程/摩擦阻力) and frictional head loss (沿程水头损失)

Frictional resistance: When the solid boundaries restricting the flow make the fluid flows uniformly, the flow resistance contains only the shear stress that is constant along the way, which is called the frictional drag.

Frictional head loss: The head loss caused by the work of frictional resistance

(2) Local resistance (局部阻力) and minor/local head loss (局部水头损失)

Local resistance: Abrupt alterations of solid boundaries cause changes of velocity distributions, further resulting in resistance, i.e. the local resistance.

Local head loss: The head loss caused by the work of local resistance

(3) Features

Friction head loss h_f : Mainly caused by frictional resistance.

Local resistance head loss h_j : Mainly due to the sudden alterations of the shapes of solid boundaries and the resulting vortex.



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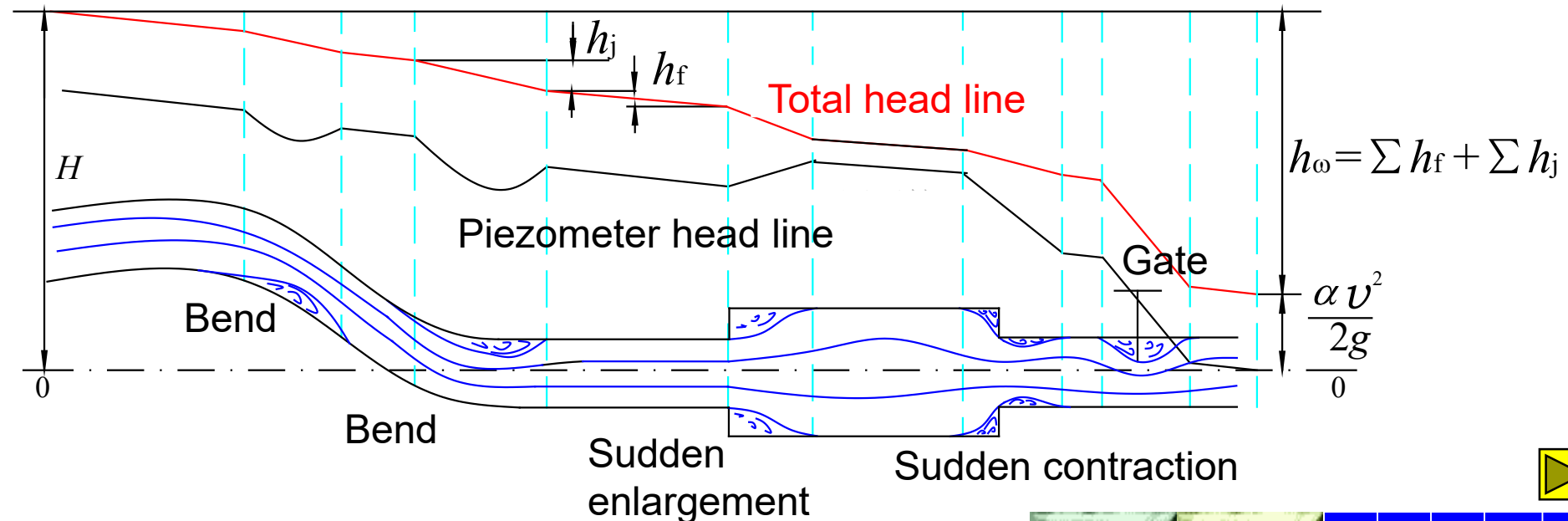
2. Basic calculation formula of head loss

(1) Principle of water head loss superposition: The head loss between two sections of a flow segment is the sum of all frictional head losses and all minor losses between the two sections. That is:

$$h_w = \sum_{i=1}^n h_{fi} + \sum_{k=1}^m h_{jk}$$

where, n ——Number of identical-section segments;

m ——Number of local-resistance forces



(2) Calculation formulation of frictional head loss

Experiments show $h_f \propto \frac{l}{d}$. Establish the following semi-empirical formula:

$$h_f = \lambda \frac{L}{d} \frac{v^2}{2g} \quad \text{which is the Darcy-Weisbach formula,}$$

where λ is the frictional loss factor.

- Scope of applications: a general formula for steady uniform pipe flow, suitable for turbulent or laminar flow in circular pipes.

For non-circular pipe: $h_f = \lambda \frac{L}{4R} \frac{v^2}{2g}$

R: hydraulic radius $R = A / \chi$

Cross section of flow

Wetted perimeter: perimeter on the flow section where fluids contact with the solid boundary

For circular pipe $R = \frac{\pi d}{4\pi d} = \frac{d}{4}$

(3) Calculation formulation of minor/local head loss

$$h_j = \zeta \frac{v^2}{2g}$$

Local resistance factor



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3. Basic equations for uniform flow (均匀流基本方程)

Force analysis on the cylindrical element between Sections 1 & 2 gives:

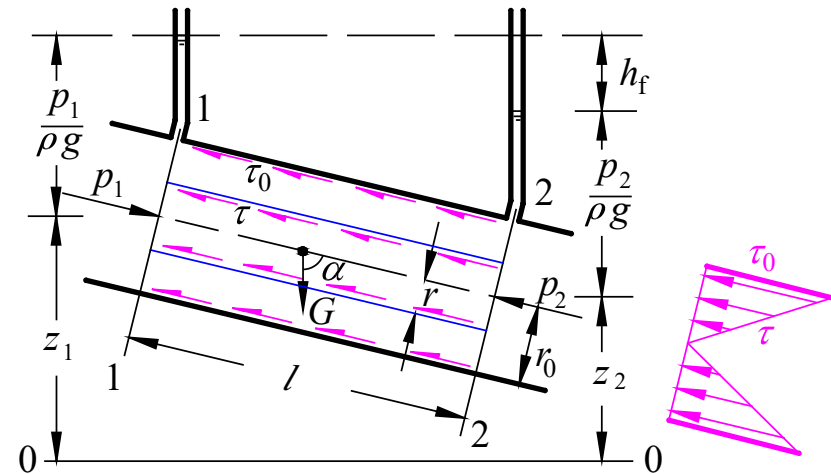
$$p_1 A' - p_2 A' + \rho g A' l \cos \alpha - \tau x' l = 0$$

Substituting $l \cos \alpha = z_1 - z_2$ and dividing by $\rho g A'$ gives:

$$\left(z_1 + \frac{p_1}{\rho g} \right) - \left(z_2 + \frac{p_2}{\rho g} \right) = \frac{\tau x' l}{\rho g A'} = \frac{\tau}{\rho g R'} l$$

$$h_{f1-2} = J \cdot l$$

$$\therefore J = \frac{\tau}{\rho g R'} \rightarrow \tau = J \rho g R'$$



$R' = R, \tau = \tau_0$ at $r = r_0$. Then we get the basic equations for uniform flow:

$$\tau_0 = \rho g R J \quad \text{or} \quad h_f = \frac{\tau_0 l}{\rho g R}$$

The equation is independent on the flow status and is applicable to both pipe flow, open channel flow, etc.

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4. Friction velocity

- Relationship between τ and λ

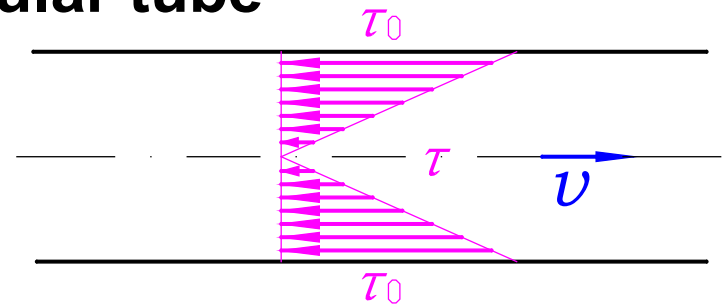
$$h_f = \lambda \frac{l}{4R} \frac{v^2}{2g} = \frac{\tau_0 l}{\rho g R} \quad \longrightarrow \quad \boxed{\tau_0 = \frac{\lambda}{8} \rho v^2}$$

- Friction velocity $v_* = \sqrt{\frac{\tau_0}{\rho}}$, with the dimension of velocity

$$v_* = v \sqrt{\frac{\lambda}{8}} = \sqrt{g R J} \quad (\text{because } \tau_0 = \rho g R J)$$

5. Shear stress distribution of circular tube

$$\frac{\tau}{\tau_0} = \frac{\rho g R' J}{\rho g R J} = \frac{R'}{R} = \frac{\frac{r}{4}}{\frac{r_0}{4}} = \frac{r}{r_0}$$



Note: Theories introduced above are independent on flow status/regime.

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-  1. Laminar flow and turbulent flow; Reynolds Experiment
- 2. Relationship between head loss and flow velocity
- 3. Identification of laminar flow and turbulent flow

Section 2 Two flow regimes of fluid motion

1. Laminar flow and turbulent flow; Reynolds Experiment

- **Laminar flow**



Laminar flow, also known as “sheet flow”, refers to that fluid elements do not mix with each other and fluid behaves as orderly stratified flow.

Features:



- 1) **Orderliness**: liquid flows in layers; fluid particles in each layer do not mix and move orderly in a straight line.
- 2) **Viscous force** dominates and follows Newton 's law of internal friction.
- 3) Energy loss is proportional to the flow velocity.
$$\tau = \mu \frac{du}{dy}$$
- 4) Occurs when the flow velocity is small and the Reynolds number Re is small.

• Turbulent Flow



Turbulent flow, also called turbulence, refers to the fluid motion, in which the local mechanical quantities such as velocity, pressure, etc. fluctuate irregularly in time and space.

Features:

1) Disorder, randomness, rotational, mixing

Fluid particles no longer flow in layers but behave irregular turbulence; particles of different layers mixed with each other, manifesting disorderly random motions and diffusion features.



2) Under the combined actions of **viscosity and turbulence**, dissipating energies

$$\tau = (\mu + \eta) \frac{du}{dy}$$

3) Head loss is proportional to the power of 1.75 ~ 2 of flow velocity.

4) Occurs when the flow velocity is large and the Reynolds number is large.

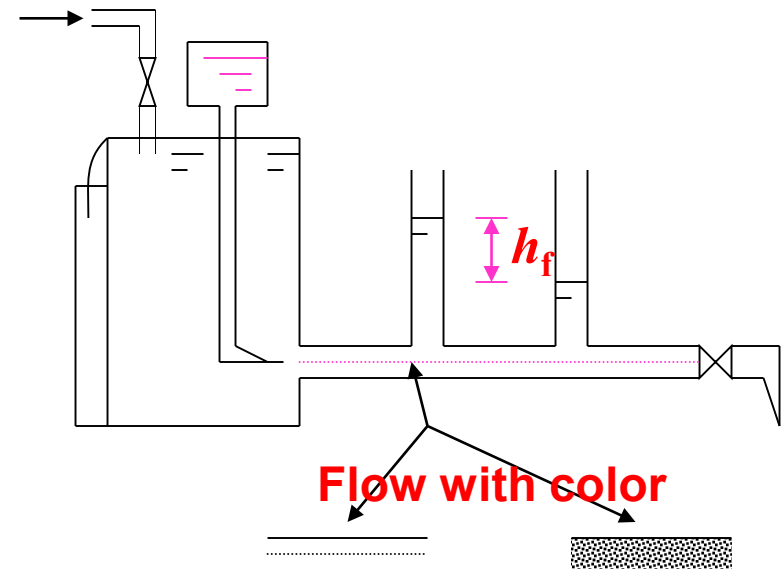
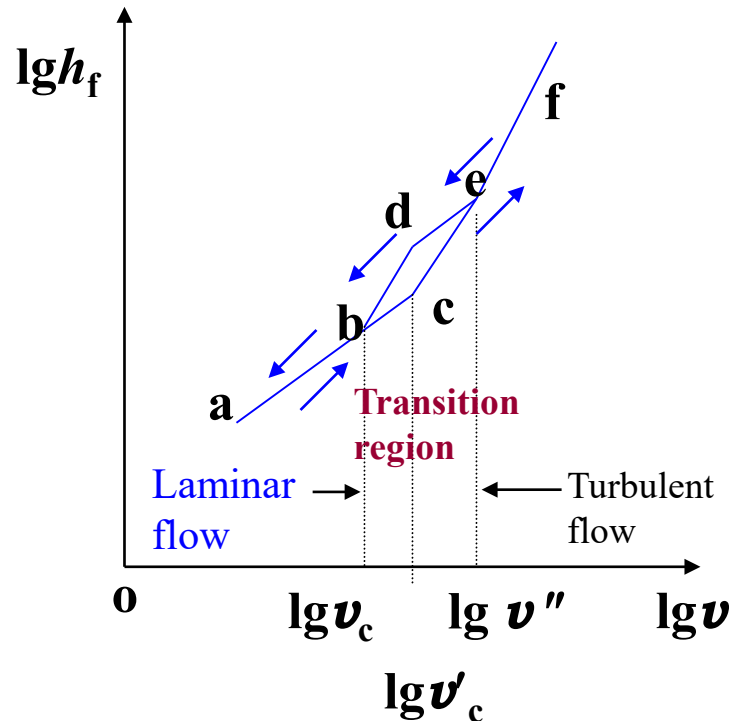


• Reynolds Experiment

In 1883, the British physicist Osborne Reynolds observed the existence of laminar and turbulent flow regimes in experiments.

The experimental curve consists of three parts:

- (1) Segment ab with $v < v_c$ corresponds to stable laminar flow
- (2) Segment ef with $v > v''$, corresponds to turbulent flow.
- (3) Segment be with $v_c < v < v''$, the flow may be laminar (bc) or turbulent (bde), depending on the previous state of the flow.



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-  2. Relationship between head loss and flow velocity
3. Identification of laminar flow and turbulent flow

2. Relationship between head Loss and flow velocity

$$\lg h_f = \lg k + m \lg v$$

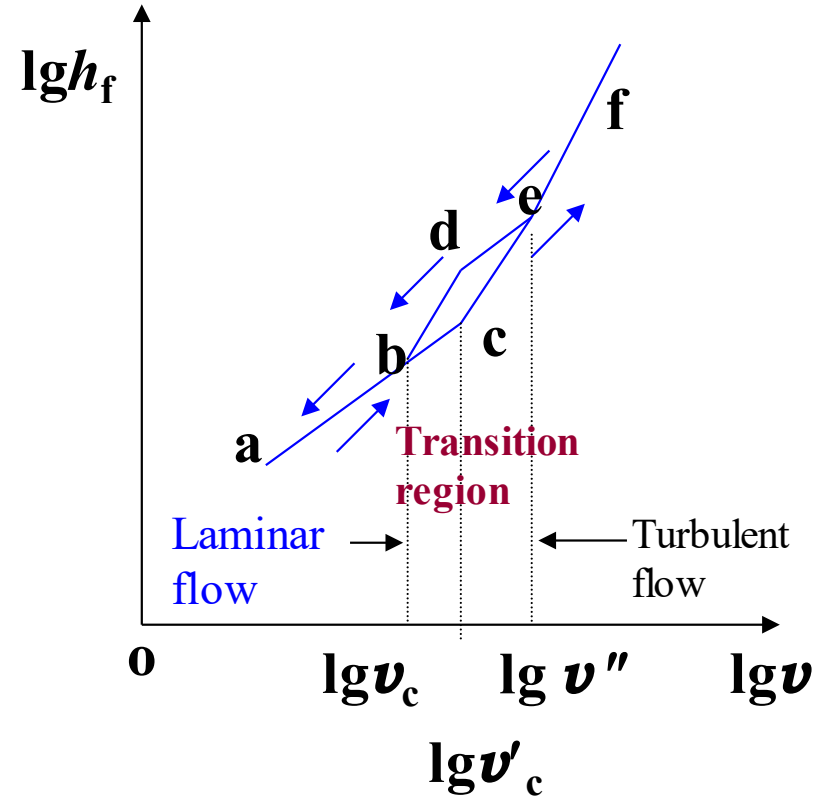
$$h_f = k v^m$$

Laminar flow: $m_1=1.0$

$$h_f = k_1 v$$

Turbulent flow: $m_2=1.75\sim 2.0$

$$h_f = k_2 v^{1.75\sim 2.0}$$



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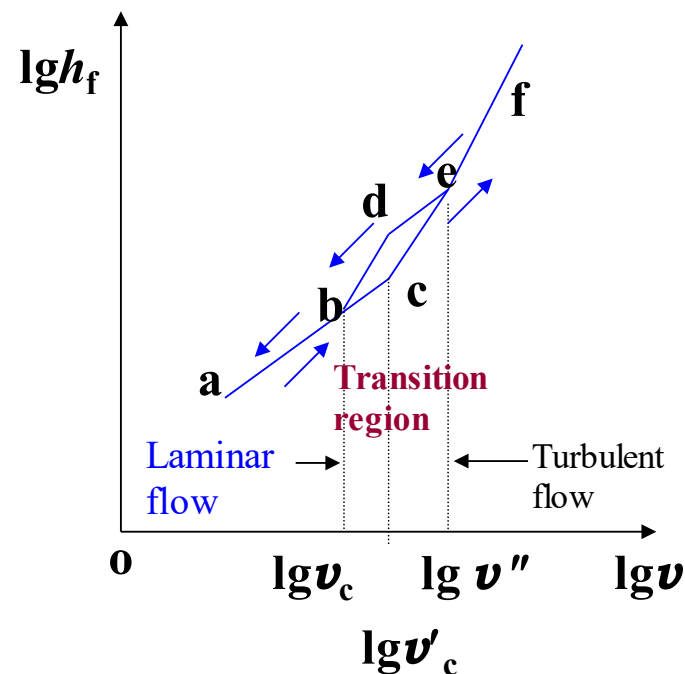
3. Identification of laminar and turbulent flows

Upper critical Reynolds number: laminar flow $\rightarrow$ turbulence

$$Re'_c = \frac{v'_c d}{\nu} \quad \text{Unstable}$$

Lower critical Reynolds number :
turbulence $\rightarrow$ laminar flow

$$Re_c = \frac{v_c d}{\nu} \quad \text{Stable}$$



• Pipe flow

$$Re_c = \frac{v_c d}{\nu} = 2300$$

$$Re < Re_c = 2300 \quad \text{Laminar flow}$$

$$Re > Re_c = 2300 \quad \text{Turbulent flow}$$

• Open-channel flow

$$Re_c = \frac{v R}{\nu} = 575$$

R : hydraulic radius, $R = A/\chi$.



Example 1: A tap-water pipe, $d=0.1\text{m}$, $v=1.0\text{m/s}$. At the water temperature of 10°C , (1) Identify the flow regime/state in the pipe; (2) In order to maintain laminar flow, what is the maximum flow velocity?

Solution: (1) At the temperature of 10°C , the kinematic viscosity of water is:

$$\nu = \frac{0.01775}{1 + 0.0337t + 0.000221t^2} = \frac{0.01775}{1.3591} = 0.0131\text{cm}^2/\text{s}$$

$$\text{Then: } Re = \frac{vd}{\nu} = \frac{100 \times 10\text{cm}}{0.0131} = 76336 > Re_c = 2300$$

Therefore, the water flow in the pipe is turbulent.

$$(2) \quad Re_c = \frac{v_c d}{\nu}$$

$$v_c = \frac{\nu Re_c}{d} = \frac{0.0131 \times 10^{-4} \times 2300}{0.1} = 0.03\text{m/s}$$

$\therefore$ To maintain laminar flow, the maximum velocity is 0.03 m/s .

Example 2: Rectangular open channel flow in a model test, bottom width $b = 0.2$ m, water depth $h = 0.1$ m, flow velocity $v = 0.12$ m/s, water temperature is 20°C . Determine the flow regime/state.

Solution: Calculating some parameters:

$$A = bh = 0.2 \times 0.1 = 0.02 \text{ m}^2$$

$$\chi = b + 2h = 0.2 + 2 \times 0.1 = 0.4 \text{ m}$$

$$R = \frac{A}{\chi} = \frac{0.02}{0.4} = 0.05 \text{ m}$$

At the temperature 20°C

$$\nu = 1.0 \times 10^{-6} \text{ m}^2/\text{s}$$

$$Re = \frac{VR}{\nu} = \frac{0.12 \times 0.05}{1.0 \times 10^{-6}} = 6000 > Re_c = 575$$

Therefore, the open channel flow is turbulent.

1. <1> $h_f = \lambda \frac{l}{4R} \frac{v^2}{2g}$ <2> $\tau_0 = \rho g R J$ <3> $\tau_0 = \frac{\lambda}{8} \rho v^2$ When can be applied?

A. Limited to laminar flow

B. Limited to turbulent flow

C. Limited to laminar and turbulent flows in pipes

☒ D. No limitations on pipe flow or open channel flow, laminar or turbulent flow

2. What factors is Reynolds number related to? What is its physical meaning? When the flow rate in a pipe is fixed, does Re increase or decrease with the increase of pipe diameter?

The Reynolds number is related to the viscosity, velocity and boundary shape of the fluid. Re implies the ratio of inertia force to viscous force.

$$Re_c = \frac{vd}{\nu} = \frac{4q_v}{\pi d\nu} \quad \text{With fixed } q_v, d \uparrow, Re \downarrow.$$

3. Why the lower critical Reynolds number rather than the upper critical Reynolds number is used as identification criterion for laminar and turbulent flows?

The upper critical Reynolds number is unstable, while the lower critical Reynolds number is stable and is only related to the shape of the flow cross section.

4. How does the lower critical Reynolds number change when the diameter of the pipe becomes larger?

Does not change, because Re_c depends only on the shape of flow boundaries, i.e., the shape of the flow cross section.

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- 1. Velocity distribution
- 2. Frictional head loss

1. Velocity distribution

Newton's law of internal friction

$$\because y = r_0 - r$$

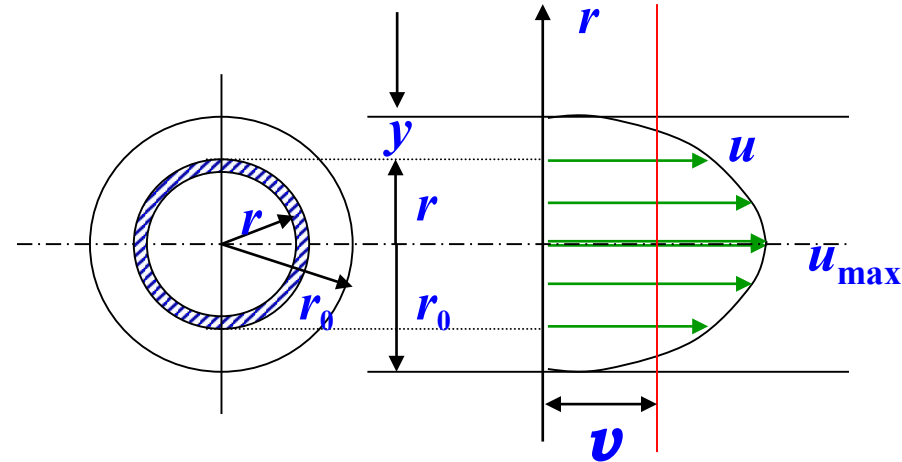
$$\frac{1}{2} r \rho g J = \tau = \mu \frac{du}{dy} = \mu \frac{du}{-dr}$$

$$du = \frac{-\rho g J}{2\mu} r dr$$

Integration gives: $u = \frac{-\rho g J}{4\mu} r^2 + C$

Substituting $u=0$ at $r=r_0$ yields: $C = \frac{\rho g J}{4\mu} r_0^2$

$$\tau = \frac{r}{r_0} \tau_0 = \frac{r}{r_0} \rho g \frac{r_0}{2} J = \frac{1}{2} r \rho g J$$



(1) Velocity distribution of laminar flow in a circular tube

$$u = \frac{\rho g J}{4\mu} (r_0^2 - r^2)$$

Physical meaning: The velocities of laminar flow on a cross section of a circular tube follows the **paraboloid of revolution** (旋转抛物面) .



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(2) Maximum flow velocity

The maximum velocity of laminar flow in a circular tube occurs at the pipe axis, i.e. $r = 0$.

$$u = \frac{\rho g J}{4\mu} (r_0^2 - r^2)$$

$$u_{\max} = \frac{\rho g J}{4\mu} r_0^2$$

Hence, $u = u_{\max} \left[1 - \left(\frac{r}{r_0} \right)^2 \right]$

(3) Mean velocity at a section

$$v = \frac{q_v}{A} = \frac{\int_A u dA}{A} = \frac{\int \frac{\rho g J}{4\mu} (r_0^2 - r^2) \cdot 2\pi r dr}{\pi r_0^2} = \frac{\rho g J}{8\mu} r_0^2 = \frac{1}{2} u_{\max}$$

The average velocity of laminar flow is half of the maximum velocity.

(4) Kinetic energy and momentum correction factors:

$$\alpha = 2.0, \quad \beta = \frac{4}{3}$$



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Section 3 Laminar flow motion in circular tube

1. Velocity distribution

👉 2. Frictional head loss (沿程损失)

2. Frictional head loss

$$\because v = \frac{1}{2} v_{\max} = \frac{\rho g J}{8\mu} r_0^2 \quad \therefore J = \frac{8\mu v}{\rho g r_0^2} = \frac{8\mu v}{\rho g (d/2)^2} = \frac{h_f}{L}$$

$$\therefore h_f = \frac{32\mu v L}{\rho g d^2} = \frac{64}{Re} \frac{L}{d} \frac{v^2}{2g} = \lambda \frac{L}{d} \frac{v^2}{2g}$$

where $\lambda = \frac{64}{Re}$ ———Frictional resistance factor for laminar flow.

Physical meaning: For laminar flows in circular pipes, the frictional head loss is proportional to **the first power** of the average flow velocity at a section and is **independent on the roughness of the pipe wall**.

Scope of applications:

- Suitable for uniform flow; invalid near the inlet of pipeline
- The velocity distribution of laminar flow is used in the derivation
- The formulation can be extended to turbulent flow. The λ value for turbulent flow is not a constant and in the form of: $\lambda = f(Re, \frac{\Delta}{d})$

Example 1: A oil with $\rho=850\text{kg/m}^3$ and $\nu=0.18\times 10^{-4}\text{m}^2/\text{s}$ moves as laminar flow in a 0.1m-diameter pipe with velocity $v=0.0635\text{ m/s}$. Solve for: (1) The maximum flow velocity at the center of pipe; (2) The flow velocity at $r = 20\text{ mm}$ from the pipe center; (3) Frictional resistance factor λ ; (4) Wall shear stress τ_0 and the head loss per 1-km pipe.

Solution: (1) Maximum flow velocity in pipe center

$$u_{\max} = 2v = 2 \times 0.0635 = 0.127\text{m/s}$$

(2) The flow velocity at $r = 20\text{mm}$ away from the pipe center

$$u = u_{\max} \left[1 - (r/r_0)^2 \right] = 0.127 \times \left[1 - (0.02/0.05)^2 \right] = 0.107\text{m/s}$$

(3) Frictional resistance factor

$$\text{Solve } Re \text{ first } Re = \frac{vd}{\nu} = 10 \times \frac{6.35}{0.18} = 353 \text{ (laminar flow)}$$

$$\text{Then } \lambda = \frac{64}{Re} = \frac{64}{353} = 0.18$$

(4) Wall shear stress and head loss per km

$$\tau_0 = \frac{\lambda}{8} \rho v^2 = \frac{0.18}{8} \times 850 \times 0.0635^2 = 0.077 \text{ N/m}^2$$

$$h_f = \lambda \frac{l}{d} \frac{v^2}{2g} = 0.18 \times \frac{1000}{0.1} \times \frac{0.0635^2}{19.60} = 0.37\text{m}$$



Example 2: Use a small-tube viscometer to measure the viscosity of oil.

The diameter of the tube is $d=6\text{mm}$; the length of the measurement section is $l=2\text{m}$. The oil flow $Q=77\times 10^{-6}\text{m}^3/\text{s}$; the reading of the mercury pressure gauge is $h_p=0.3\text{m}$; the oil density is 900kg/m^3 . Solve the kinematic viscosity and dynamic viscosity of oil.

Solution: Bernoulli equation of front and rear section of split thin tube

$$h_f = \frac{p_1}{\rho g} - \frac{p_2}{\rho g} = \left(\frac{\rho_p}{\rho} - 1\right)h_p = \left(\frac{13600}{900} - 1\right) \times 0.3 = 4.23\text{m}$$

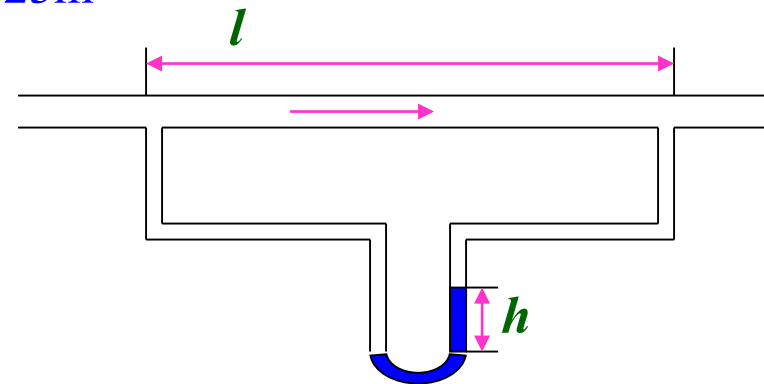
Assume the flow to be laminar

$$v = \frac{4q_v}{\pi d^2} = 2.73\text{m/s}$$

$$h_f = \frac{64\nu}{vd} \cdot \frac{l}{d} \cdot \frac{v^2}{2g}$$

$$\nu = h_f \frac{2gd^2}{64lv} = 4.23 \times \frac{2g \times (0.006)^2}{64 \times 2 \times 2.73} = 8.54 \times 10^{-6}\text{m}^2/\text{s}$$

$$\mu = \rho\nu = 900 \times 8.54 \times 10^{-6} = 7.69 \times 10^{-3}\text{Pa}\cdot\text{s}$$



Check condition $Re = \frac{vd}{\nu} = \frac{2.73 \times 0.006}{8.54 \times 10^{-6}} = 1918 < 2300$

Laminar flow

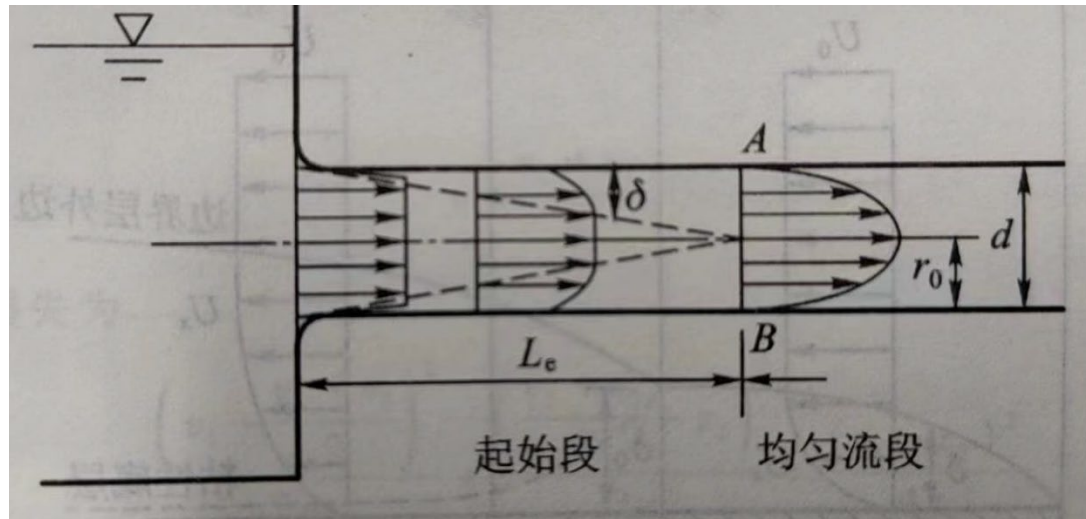
1. How are the distributions of the shear stress and velocity of laminar flow in circular tube?

- Shear stress: Linear distribution; the value at the pipe center is 0 & the maximum occurs at the surface of the circular pipe wall;
- Velocity: Distribution as paraboloid of revolution (回转抛物面); the maximum at the pipe center and zero on the pipe wall.

2. How to calculate the frictional resistance factor for laminar flow in circular pipes? Is the formula applicable for the inlet of circular pipes? Why?

$$\lambda = \frac{64}{Re}$$

No, because the velocity is not paraboloid of revolution.



3. Why the flow velocity closing to the pipe wall near the inlet section of the pipe decreases gradually, while that at the pipe center increases gradually?

- The velocity distribution on the section near the inlet is generally uniform, and the velocity gradient exists mainly at regions very close to the pipe wall. Hence, the shear stress near the wall is large, and so does the flow resistance, which causes the gradual decrease of flow velocity.
- The velocity gradient near the pipe center is small $\rightarrow \tau$ and the flow resistance are small. Hence the flow velocity increases. Until a certain velocity gradient and shear stress are formed, the energy consumption and energy supplement of each part of the fluid are balanced.

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